

SYNAPSE AI

Complete Feature Expansion Roadmap

AETHEX Medical Platform • Built for Doctors & Medical Students

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New Features	Mode Expansions	AI Upgrades	Monetization

Clinical Tools

Drug Interactions Checker	Real-time detection of drug-drug, drug-food interactions. Auto-flag severity (mild/moderate/severe) with clinical alternatives.	High Priority
Dosage Calculator	Weight-based dosing, renal/hepatic dose adjustments. Supports paediatric, geriatric, and ICU protocols.	High Priority
Differential Diagnosis Generator	Input symptoms + vitals → ranked DDx list with ICD-10 codes, red flags, and suggested workup plan.	High Priority
Lab Values Interpreter	Paste CBC, LFT, KFT, ABG, lipid panel results → instant AI interpretation with clinical significance and action points.	Medium Priority

Mode Expansions (extending Diagnose / Research / Image / Slides)

Case Summary — SOAP Notes	Auto-generate structured Subjective / Objective / Assessment / Plan notes from a clinical narrative in seconds.	High Priority
Patient Education Mode	Translate complex diagnoses into simple, layperson language. Supports Hindi/regional language output for Indian patients.	Medium Priority
Procedure Guide	Step-by-step clinical procedure walkthroughs — LP, central line, intubation, suturing — with annotated images.	Medium Priority
MCQ Generator	Generate USMLE/NEET-PG style questions from any topic. Ideal for medical students and exam prep sessions.	Medium Priority

AI Enhancements

Voice Input (Web Speech API)	Hands-free clinical queries — doctors can speak their question without interrupting patient interaction.	High Priority
Session History & Saved Cases	Persistent per-user chat history. Save, tag, and revisit past diagnoses and research sessions.	Medium Priority
Specialty Filter	Route queries through specialty-specific AI context: Cardiology, Neurology, Paediatrics, Obstetrics, Oncology, etc.	Medium Priority

Citation Sources per Response	Every research/diagnosis answer surfaces PubMed / UpToDate / WHO guideline citations inline for credibility.	High Priority
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■ Monetization / Pro Features

Medical Image Analysis	Upload X-ray, ECG, fundus photo, MRI slice → AI-assisted read with confidence score. PRO-tier feature (SYNAPSE Pro).	High Priority
PDF Export	Download any diagnosis, research output, or SOAP note as a branded AETHEX PDF — shareable with patients/colleagues.	Medium Priority
EMR-Style Case File Builder	Build a structured patient case file inside SYNAPSE — demographics, history, investigations, diagnosis, plan. Exportable.	Low Priority

■ UX / Interface Improvements

Dark / Light Mode Toggle	User preference for theme. Dark is current default; light mode improves readability in clinical settings.	Low Priority
Pinned & Favourite Prompts	Let users save frequently used queries (e.g. 'ACLS protocol', 'sepsis management') for one-click access.	Low Priority
Mobile-Responsive Layout	Fully optimised for mobile browsers — critical for doctors using phones during rounds or emergencies.	High Priority

■ Recommended Implementation Stack

Layer	Technology	Purpose
Frontend	React + Tailwind CSS	Responsive UI, voice input, mode tabs
AI Core	Claude API (Anthropic)	Diagnosis, DDx, research, MCQ generation
Image Analysis	GPT-4o Vision / Gemini Pro Vision	X-ray, ECG, fundus AI read
Voice Input	Web Speech API	Hands-free clinical queries
Backend	FastAPI (Python)	API gateway, session management, auth
Database	PostgreSQL + Redis	User sessions, case history, caching
PDF Export	ReportLab	Branded AETHEX PDF generation
Deployment	Replit → Vercel (aethex.in)	CI/CD, serverless functions